

The Role of Solitons in the Energy and Electron Transfer in One-Dimensional Molecular Systems

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Abstract

Theory of motion of quasi-particles, electrons and a vibrational excitation along molecular chains is developed without using the adiabatic approximation and the perturbation theory. It is shown that the motion of quasi-particles is accompanied by a local chain deformation. The deceleration of a soliton by external dissipative forces is studied. The effect of thermal vibrations on the soliton motion is analysed.

1. Introduction

When studying biological phenomena on a molecular level it was established that many processes are associated with an electron transfer along protein molecules from donor to acceptor molecules or with a transfer of energy released under hydrolysis of adenosine triphosphate (ATP) molecules.

The energy of ATP molecule hydrolysis equal to 0.54 eV is only 20 times greater than the average energy to thermal vibrations. Many researchs suggested that the energy of the ATP hydrolysis moves along protein molecules as vibrational oscillations of atoms $C = O$ contained in peptide groups of protein molecules. However, at standard conditions the life time of such vibrations is only 10^{-13} s. Therefore many doubts have been expressed concerning the possibility of an effective transfer of vibrational energy along protein molecules.

Great difficulties are also experienced in the explanation of the effective transfer from donor to acceptor molecules. The experimental data show that such a transfer is possible the distance of the order of $70 \text{ \AA}$. The electron transfer to such great distances cannot be explained by the usual theory of quantum mechanical tunneling.

One of possible explanations of the electron transfer at large distances is connected with the assumption that the transfer process is facilitated by protein molecules to which a donor and acceptor of electrons are attached. The donor-acceptor model of the electron transfer through protein molecules was developed in [1-2]. In these papers a protein molecule was modelling by a quasi-one-dimensional molecular system which contains one extra electron passed into a molecule from a donor. In these and in most other papers in which the motion of quasi-particles (electrons, excitons etc.) was studied in periodic structures an adiabatic approximation was used. When using such an approximation one assumes that molecules slightly displace with respect to stable equilibrium positions. The interaction of a quasi-particle with such displacements is considered as a perturbation decelerating its motion.

Recently in our institute the study has been made of the motion of quasi-particles - electrons and vibrational excitation - along molecular chains without using the adiabatic approximation and the perturbation theory. Giving the perturbation theory up leads to a description of quasi-particle motion by means of nonlinear equations. Their solution shows [3-9] that the motion of a quasi-particle is accompanied by a local deformation of a chain, i.e. a displacement of equilibrium positions. The simultaneous motion of a quasi-particle and a local deformation is described by a wave function in the form of a solitary wave-soliton - which is a solution of a nonlinear Schrodinger equation.

Solitons transport a dipole vibrational excitation or an electron with a local decrease of mutual distances between peptide groups. These solitons always travel with a velocity less than that of a sound, therefore, they do not expend energy to generate phonons. The energy expenditure is required to break the electron coupling or vibrational excitations of peptide groups with a local deformation of a molecule. Thus, the motion of an electron or a vibrational excitation of peptide groups along a protein molecule holds without energy loss and resembles superconductivity and superfluidity phenomena.

Below we consider the basic results of theoretical investigations [3-9].

2. Solitons in a One-Dimensional Molecular System at Very Low Temperatures

We consider a one-dimensional molecular chain with repetitive peptide groups as a simplest protein molecule. As a result of a resonant interaction of vibrational excitations of peptide groups, in such a chain the collective states appear which may be described by the Hamiltonian

$$H = H_e + H_{ph} + H_{int} \quad (2.1)$$

where

$$H_e = \sum_n \left(\mathcal{E}_0 - \frac{\hbar^2}{m a^2} \right) B_n^+ B_n + \frac{\hbar^2}{2 m a^2} B_n^+ (B_{n-1} + B_{n+1}) \quad (2.2)$$

is the Hamiltonian of collective excitations - excitons in the chain with peptide groups fixed rigidly in the equilibrium positions (at the points na); excitons are described by plane waves. a is the distance between the nearest points, m is the effective exciton mass; $\mathcal{E}_0$ is the energy of the exciton band bottom; B_n^+ and B_n are the creation and annihilation operators of the vibrational excitation on the group of the number n .

$$H_{ph} = \frac{1}{2} \sum_n \left(\frac{1}{M} \hat{p}_n^2 + \mathcal{A} u_n^2 \right) \quad (2.3)$$

is the operator of longitudinal displacements u_n in peptide groups of the mass M from equilibrium positions, $\mathcal{A}$ is the viscosity coefficient, $\hat{p}_n$ is the momentum operator.

$$H_{int} = \chi \sum_n (u_{n+1} - u_{n-1}) B_n^+ B_n \quad (2.4)$$

is the operator of interaction of vibrational excitations with displacements of peptide groups. χ is the parameter of this interaction.

The operator (2.1) may also describe the motion of an extra electron in a molecular chain. In this case the operator (2.2) will characterize the extra electron motion in repetitive potential

wells caused by constant electric dipole moments of peptide groups. Then m is the effective electron mass, $\mathcal{E}_0$ is the energy of the conductivity band bottom ^{*)}.

Let us determine the Hamiltonian function in our system by the equality

$$\mathcal{H} = \langle \Psi(t) | H | \Psi(t) \rangle$$

where the function

$$\Psi(t) = \sum_n \varphi_n(t) U(t) B_n^\dagger |0\rangle, \quad \sum_n |\varphi_n|^2 = 1 \quad (2.5)$$

contains the unitary operator of displacements

$$U(t) = \exp \left\{ -\frac{i}{\hbar} \sum_n \left[\beta_n(t) \hat{p}_n - \pi_n(t) u_n \right] \right\}. \quad (2.6)$$

The Hamiltonian function $\mathcal{H}$ depends on $\varphi_n(t)$, $\beta_n(t)$ and on their canonically conjugate $i\hbar \varphi_n^*(t)$ and $\pi_n(t)$. In this case $|\varphi_n(t)|^2$ determines the possibility of vibrational excitation of the group n ; $\beta_n(t)$ characterizes the displacement of equilibrium positions of peptide groups.

Using the Hamiltonian function $\mathcal{H}$ we may find the Hamiltonian equation

$$i\hbar \frac{\partial \varphi_n}{\partial t} = \frac{\partial \mathcal{H}}{\partial \varphi_n^*}, \quad \frac{\partial \beta_n}{\partial t} = \frac{\partial \mathcal{H}}{\partial \pi_n}, \quad \frac{\partial \pi_n}{\partial t} = -\frac{\partial \mathcal{H}}{\partial \beta_n}.$$

In the continuum approximation these equations are reduced to two equations

$$\left[i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\chi \rho(x,t) \right] \varphi(x,t) = 0, \quad (2.7)$$

^{*)} Below we shall speak only about the motion of vibrational excitation. However all the results equally refer to the electron motion. It is sufficient to replace the word exciton by the word electron.

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2}\right) \rho(x,t) + \frac{2\chi a^2}{M} \frac{\partial^2}{\partial x^2} |\varphi(x,t)|^2 = 0 \quad (2.8)$$

where $V_{aq} = a\sqrt{\frac{e}{M}}$ is the velocity of the longitudinal sound in the chain

$$\rho(x,t) = -a \frac{\partial \beta(x,t)}{\partial x} \quad (2.9)$$

is the relative decrease in distances between neighbouring peptide groups.

Owing to translational invariance of the chain we may seek solution of the set of equations (2.7) and (2.8) in the form of excitations spreading with a constant velocity V . Assuming

$$\text{where } \rho(x,t) = \rho(\xi) \quad , \quad |\varphi(x,t)|^2 = f(\xi) \\ \xi \equiv x - x_0 - Vt \quad (2.10)$$

we find the equation

$$\rho(\xi) = \frac{2\chi}{ae(1-s^2)} |\varphi(x,t)|^2 \quad , \quad s \equiv V/V_{aq} \quad (2.11)$$

from Eq.(2.8) integrating twice over the variable ξ . Substituting (2.11) into the equation (2.8) we obtain the nonlinear Schrödinger equation

$$\left(i\hbar \frac{\partial}{\partial t} - \mathcal{E}_0 + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + G(s) |\varphi(x,t)|^2\right) \varphi(x,t) = 0 \quad (2.12)$$

with the nonlinearity parameter

$$G(s) = \frac{G(0)}{1-s^2} \quad G(0) \equiv 4\chi^2/ae \quad (2.13)$$

At $s > 1$ the value $G(s) < 0$, therefore the solution normalized by the condition

$$\frac{1}{a} \int |\varphi(x,t)|^2 dx = 1 \quad (2.14)$$

is absent in an infinite chain. In a chain of a finite length L ($L \gg a$) the solution of Eq.(2.12) is expressed by a plane wave

$$\varphi(x,t) = L^{-1/2} \exp \{ i [kx - \omega(k)t] \} \quad (2.15)$$

and corresponds to the energy

$$\hbar \omega(k) = \varepsilon_0 + \frac{a}{L} |G(s)| + \frac{\hbar^2 k^2}{2m}, \quad G(s) < 0, \quad (2.15a)$$

where $k = mV/\hbar$. Thus, at $s > 1$ the collective excitations in the chain are excitons.

With velocities V less than that of longitudinal sound V_{aq} the value $G(s) > 0$. In this case the equation (2.12) has the following solution normalized by the condition (2.14)

$$\varphi(x,t) = \frac{\sqrt{a\mu} \exp \{ i [kz - (\omega - kV)t] \}}{\sqrt{2} \cosh(\mu z)}, \quad (2.16)$$

$$\mu = \frac{amG(s)}{2\hbar^2} = \frac{2am\chi^2}{2e\hbar^2(1-s^2)}, \quad (2.17)$$

$$\hbar \omega = \varepsilon_0 + \frac{\hbar^2(k^2 - \mu^2)}{2m}, \quad k = mV/\hbar. \quad (2.18)$$

The excitations described by the function (2.16) are called solitons. The value $\Delta x = \hbar/\mu$ characterizes the region of the excitation localization (the soliton size). The continuum approximation used is justified when the inequality $\Delta x \gg a$ is satisfied. With fixed parameters of the chain (e, χ, m) this inequality imposes a restriction on the soliton motion velocity.

According to (2.11) the decrease in the distances between peptide groups is determined by the expression

$$\rho(\zeta) = \frac{a m \chi}{(1-s^2) \alpha \cosh^2(m\zeta)}. \quad (2.19)$$

The displacement of equilibrium positions of molecules is expressed by the function

$$\beta(\zeta) = \frac{\chi}{\alpha(1-s^2)} [1 - \tanh(m\zeta)]. \quad (2.19a)$$

The soliton energy travelling with velocity V involves also the energy of local deformation. It is determined by the expression

$$E(V) = \mathcal{E}_0 + \frac{1}{2} m V^2 - \frac{4ma^2\chi^4(1-3s^2)}{3\alpha^2\hbar^2(1-s^2)^2}$$

At $s \ll 1$ this expression is transformed to the form

$$E(V) = \mathcal{E}_0 + \frac{1}{2} m_{\text{sol}} V^2 - \frac{4ma^2\chi^4}{3\alpha^2\hbar^2}, \quad (2.20)$$

where

$$m_{\text{sol}} = m \left(1 + \frac{8M\chi^4}{3\alpha^3} \right). \quad (2.21)$$

In the chain of the finite length L the equation (2.12) permits also solutions of the exciton type described by plane waves (2.15) with the energy

$$\hbar\omega(k) = \mathcal{E}_0 - \frac{aG(s)}{L} + \frac{\hbar^2 k^2}{2m}, \quad k = mV/\hbar. \quad (2.22)$$

At $a \ll L$ the inequality

$$\frac{aG(s)}{L} \ll \frac{4ma^2\chi^4}{3\alpha^2\hbar^2} \quad (2.23)$$

is usually satisfied, therefore, the internal soliton energy (at $V = 0$) is less than the sum of the internal exciton and deformation energy. Thus, the exciton excitations are unstable.

The exciton motion is not practically accompanied by the chain deformation. The exciton transport only the vibrational excitation energy. The soliton motion corresponds to the vibrational excitation motion in conjunction with a local deformation. Therefore, the effective soliton mass (m_{sol}) is greater than the effective exciton mass (m). This difference is especially great in soft chains (small κ) and at a strong coupling of a vibrational excitation with deformation (large χ). Travelling with a velocity less than that of a sound the solitons expend no energy to generate phonons.

The excitons are excited by light in protein molecules. The soliton excitation is forbidden by the Frank-Condon principle according to which an optical transfer should not be accompanied by displacement of heavy particles. The solitons may be excited only under local external influences for example, under chemical reactions occurring with molecules attached to the end of protein molecule.

The simple model of a quasi-one-dimensional periodic structure under consideration is referred to each of three chains formed in an α -helical protein molecule by hydrogen bonds between peptide groups. In studying the electron motion each chain acts independently. When vibrational excitations of the dipole type are investigated an additional resonance interaction appears between the peptide groups contained in different chains. Therefore, all three chains should be considered as a whole. This problem was solved in [4].

In the paper of Hymen, Mc Layhlin and Scott [14] the collective excitations in alpha-helical protein molecules of the finite length ($L = 200a$) were computed without making use of the continuum approximation. The conditions of soliton excitation and spreading in such molecules were elucidated. Authors of [14] say that numerical studies of dynamic model for α -helix protein confirm the prediction in [4] of soliton formation.

3. Deceleration of Solitons by external dissipative forces

The possibility of making use of the motion of soliton to explain some phenomena in physics and biology is inevitably connected with studying the physical conditions leading to nonlinearity and dispersion. The properties of solitons are manifested under their interaction with external fields and other degrees of freedom of the system. Mathematical study of these interactions is usually carried out by the perturbation theory methods. Different variants of the perturbation theory for solitons in one-dimensional systems were developed by Karpman and Maslov [10], Bishop and Krumhanse et al [11], Kinner and Mc Laughlin [12] and by other investigators.

In this section we give the results of investigations [8] concerning the influence of external effects on solitons described by the set of equations (2.7), (2.8). The nonlinearity of the collective excitations described by these equations is determined by the interaction of an exciton (electron) with a local chain deformation generated by this exciton. Below we shall consider only the solitons moving with velocity V significantly less than that of sound waves V_{aq} .

If a molecular chain is in a certain medium, then a moving soliton will be acted upon by a friction force and it will be decelerated. The friction forces decelerate the displacements of peptide groups. We consider a case when the friction forces are proportional to the velocity of displacements $\sim \gamma \frac{\partial \beta}{\partial t}$. In this case the equations (2.8) determining the relative displacement $\rho(x, t)$ are transformed to the form (see [8]):

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2} \right) \rho(x, t) + \frac{2\chi a V_{aq}^2}{\partial x} \frac{\partial^2}{\partial x^2} |\varphi(x, t)|^2 = -\varepsilon \gamma \frac{\partial \rho(x, t)}{\partial t}. \quad (3.1)$$

For convenience, a small positive parameter $\varepsilon \ll 1$ is introduced into the deceleration coefficient $\varepsilon \gamma$. We introduce the transformation

$$\varphi(x, t) = \Phi(x, t) \exp(-\varepsilon_0 t / \ell)$$

then according to (2.7) the function $\Phi(x, t)$ will satisfy the equation

$$\left[i \hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\chi \rho(x,t) \right] \Phi(x,t) = 0. \quad (3.2)$$

In order to solve the set of equations (3.1), (3.2) we use the method of slowly varying coefficients introduced by Van der-Pole and developed in detail by Bogoliubov and Mitropolskii [13].

Assume

$$\Phi(x,t) = \Phi_0(x,t) + \varepsilon \Phi_1(x,t), \quad (3.3)$$

$$\rho(x,t) = \rho_0(x,t) + \varepsilon \rho_1(x,t), \quad (3.4)$$

where

$$\Phi_0(x,t) = \frac{\sqrt{am} \exp \{ i [k(x-\xi) + \Theta] \}}{\sqrt{2} \cosh [\mu(x-\xi)]} \quad (3.5)$$

$$\xi = x_0 + \frac{\hbar k}{m} t, \quad \mu = \frac{2am\chi^2}{2e\hbar^2}, \quad (3.5a)$$

$$\rho_0(x,t) = \frac{2\chi}{2e} |\Phi(x,t)|^2. \quad (3.6)$$

The function (3.5) describes the soliton travelling with velocity $V = \hbar k/m$. The value

$$x = \xi \equiv x_0 + \frac{\hbar k}{m} t \quad (3.7)$$

determines the position of the soliton centre at the time moment t :

$$\Theta = \hbar(k^2 + \mu^2)t/2m \quad (3.8)$$

is the phase of the wave function (3.5).

The presence of friction results in gradual changing the parameters k and Θ of the function (3.5) and in a small changing the form of the function (3.5) which is taken into account by the term $\varepsilon \Phi_1$ in (3.3). We express these changes by the unknown functions $f(k)$ and $\varphi(k)$ using the equations

$$\frac{dk}{dt} = \varepsilon f(k), \quad (3.9)$$

$$\frac{d\Theta}{dt} = \frac{\hbar^2}{2m} (k^2 + \mu^2) + \varepsilon \varphi(k), \quad (3.9a)$$

$$\frac{d^2 \xi}{dt^2} = \varepsilon \hbar f(k) / m. \quad (3.9b)$$

Substituting (3.4) into Eq.(3.1) and taking into account (3.5) and (3.6) and a smooth dependence $\rho_1(x, t)$ on time we obtain

$$\rho_1(x, t) = - \frac{a \chi \hbar}{2em V_{a_1}^2} \{ \delta k^2 + f(k) \} \tanh [\mu (x - \xi)]. \quad (3.10)$$

Using (3.3) and (3.4) and the equalities

$$\frac{\partial \Phi_0}{\partial k} = i(x - \xi) \Phi_0, \quad \frac{\partial \Phi_0}{\partial \Theta} = i \Phi_0$$

we may transform the equation (3.2) to the form

$$\left[i \hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{8\chi^2}{\varepsilon} |\Phi_0|^2 \right] \Phi_1 + \frac{4\chi^2}{\varepsilon} \Phi_0^2 \Phi_1^* = S(\Phi_0), \quad (3.11)$$

where

$$S(\Phi_0) \equiv \left[(x - \xi) f(k) + \varphi(k) - 2\chi \rho_1(x, t) \right] \Phi_0. \quad (3.12)$$

The unknown functions $f(k)$ and $\varphi(k)$ introduced into (3.9) are determined provided that the function Φ_1 in (3.11) has no secular terms increasing quadratically with time. One may show [8] that

such secular terms contain multipliers $\frac{\partial \Phi_0}{\partial \xi}$ and $\frac{\partial \Phi_0}{\partial \Theta}$. Therefore, the condition that Φ_1 involves no secular terms is reduced to the requirement of orthogonality of the functions $\frac{\partial \Phi_0}{\partial \xi}$ and $\frac{\partial \Phi_0}{\partial \Theta}$ in the right-hand side of (3.11) i.e. to the fulfillment of the equalities

$$\int S^*(\Phi_0) \frac{\partial \Phi_0}{\partial \xi} dx = \int S^*(\Phi_0) \frac{\partial \Phi_0}{\partial \Theta} dx = 0. \quad (3.13)$$

When these equalities are satisfied only virtual exciton states contribute into $\Phi_1(x, t)$. They cause the soliton form distortion and the so called "tail outgrowing" [10].

Since the energy of exciton states exceeds that of a soliton by the value $ma^2\chi^4/\epsilon^2\hbar^2$, then at large χ and small ϵ the value $B_1(x, t)$ will be very small for all times t less than $k_0/\epsilon f(k_0)$. Therefore, the explicit form of the function $B_1(x, t)$ is not calculated.

The inequalities (3.13) yield the explicit expressions of the unknown functions

$$\begin{aligned} f(k) &= \frac{4m\chi}{\hbar} \int \rho_1(x, t) |\Phi_0(x, t)|^2 \tanh[\mu(x-\xi)] dx, \\ \varphi(k) &= \frac{2\chi}{a\hbar} \int \rho_1(x, t) |\Phi_0(x, t)|^2 dx. \end{aligned} \quad (3.14)$$

Substituting ρ_1 and Φ_0 into (3.14) we obtain

$$\varphi(k) = 0, \quad f(k) = - \frac{\delta D k}{1+D} \quad (3.15)$$

where

$$D = \frac{8a^2\chi^4}{3\epsilon^2\hbar^2 V_{aq}^2} = \frac{m_{sol} - m}{m} > 0. \quad (3.16)$$

Substituting (3.15) into (3.9) and integrating over time we obtain the exponential law of decreasing $k(t)$ and hence of decreasing

soliton velocity

$$V(t) = V_0 \exp(-\alpha t) \quad (3.17)$$

where the damping coefficient due to overcoming the friction forces

$$\alpha = \varepsilon \gamma \frac{m_{sol} - m}{m_{sol}}. \quad (3.17a)$$

If the friction force is proportional to the square of velocity and has an opposite direction, i.e. is proportional to $\varepsilon \lambda \left(\frac{\partial \Phi}{\partial t}\right)^2$, then when the soliton moves along the positive direction of the axis x the equation (2.8) is replaced by the equation

$$\left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2}\right) \rho(x, t) + \frac{2\lambda V_{aq}^2}{\partial x} \frac{\partial^2}{\partial x^2} |\varphi(x, t)|^2 = -\varepsilon \lambda \frac{V}{a} \frac{\partial}{\partial t} [\rho(x, t)]^2.$$

Assuming

$$\Phi(x, t) = \Phi_0(x, t) + \varepsilon \Phi_1^1(x, t),$$

$$\rho(x, t) = \rho_0(x, t) + \varepsilon \rho_1(x, t),$$

where $\Phi_0(x, t)$ and $\rho_0(x, t)$ are determined by the expressions (3.5) and (3.6), respectively, we obtain

$$\begin{aligned} \rho_1^1(x, t) = & -\frac{a\lambda\hbar f(k)}{m\partial x V_{aq}^2} \tanh[\mu(x-\xi)] - \\ & -\frac{\lambda V_{aq}^2 a\lambda^2 \mu}{\partial x^2 V_{aq}^2} \left\{ \tanh[\mu(x-\xi)] - \frac{1}{3} \tanh^2[\mu(x-\xi)] \right\}. \end{aligned} \quad (3.18)$$

The unknown functions of Eqs (3.9) are now determined by the expressions

$$\varphi(k)=0, \quad f(k) = \frac{4\mu\lambda}{\hbar} \int \rho_1^1(x, t) \tanh[\mu(x-\xi)] |\Phi_0|^2 dx. \quad (3.19)$$

Substituting (3.18) into (3.19) we obtain the final expression

$$f(k) = - \frac{\lambda A}{1+D} V^2 \quad (3.20)$$

where D is determined by the formula (3.16) and

$$A = \frac{16 a m^2 \chi^3}{15 \varepsilon^2 \hbar V_{aq}^2}.$$

Using (3.20) we transform the equation (3.9b) to the form

$$\frac{dV}{dt} = -\delta V^2, \quad \delta = \frac{\varepsilon \lambda 16 a m^2 \chi^3}{15 m_{sol} \varepsilon^2 V_{aq}^2}, \quad (3.21)$$

Integrating (3.21) we obtain the final expression of the law of the soliton velocity change in the presence of friction proportional to the square of velocity

$$V(t) = \frac{V_0}{1 + V_0 \delta t} \approx V_0 (1 - V_0 \delta t), \quad (3.22)$$

Thus, with friction proportional to the square of velocity, the soliton velocity decreases with time by the law close to that of linear with the proportionality coefficient δV_0^2 .

4. Motion of a soliton (electron) in the presence of external nonhomogeneous fields

The presence of external and small local nonmonogeneities in the arrangement of molecules may be taken into account by introducing in (3.2) the small potential energy $\varepsilon U(x)$ depending on the coordinate x . In this case the set of equations determining the soliton motion with velocity $V \ll V_{aq}$ takes the form

$$\left[i \hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 2\chi \rho(x,t) \right] \Phi(x) = \varepsilon U(x) \Phi(x), \quad (4.1)$$

$$\left[\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2} \right] \rho(x,t) + \frac{2\chi}{\varepsilon} V_{aq}^2 \frac{\partial^2}{\partial x^2} |\Phi(x,t)|^2 = 0. \quad (4.2)$$

We may seek the solution of this set of equation in the first approximation in the form

$$\Phi(x,t) = \Phi_0(x,t) + \varepsilon \Phi_1(x,t), \quad (4.3)$$

$$\rho(x,t) = \rho_0(x,t) + \varepsilon \rho_1(x,t), \quad (4.4)$$

where

$$\rho_0(x,t) = \frac{2\chi}{\varepsilon} |\Phi_0(x,t)|^2. \quad (4.5)$$

The function Φ_0 has the form (3.5) with parameters k , ξ , ω which vary with time according to Eqs (3.9).

Substituting (4.4) into Eqs (4.1) and (4.2) we obtain

$$\begin{aligned} \left[i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{4\chi^2}{\varepsilon} |\Phi(x,t)|^2 \right] \Phi(x,t) &= \varepsilon [U(x) + \rho_1(x,t)] \Phi(x,t), \quad (4.6) \\ \left(\frac{\partial^2}{\partial t^2} - V_{aq}^2 \frac{\partial^2}{\partial x^2} \right) \rho_1(x,t) &= - \frac{1}{\varepsilon} \frac{\partial \rho_0(x,t)}{\partial \xi} \frac{d^2 \xi}{dt^2}. \end{aligned} \quad (4.7)$$

Using the value (3.9b) and a weak dependence $\rho_1(x,t)$ on time we obtain (at $S \ll 1$) the equation

$$\frac{\partial^2 \rho_1}{\partial x^2} = \frac{2a\hbar\chi\mu^2}{m\alpha\varepsilon V_{aq}^2} f(k) \frac{\sinh[\mu(x-\xi)]}{\cosh^3[\mu(x-\xi)]}.$$

After integration over x we have

$$\rho_1(x,t) = - \frac{a\hbar\chi}{m\alpha\varepsilon V_{aq}^2} f(k) \tanh[\mu(x-\xi)]. \quad (4.8)$$

To obtain the equation determining the correction $\Phi_1(x,t)$ to the function $\Phi_0(x,t)$, we substitute (4.3) into (4.6). Then we obtain

$$\begin{aligned} & \left(i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{8\chi^2}{\partial e} |\Phi_0(x,t)|^2 \right) \Phi_1(x,t) + \\ & + \frac{4\chi^2}{\partial e} \Phi_0^2(x,t) \Phi_1^*(x,t) = S(\Phi_0), \end{aligned} \quad (4.9)$$

where

$$\begin{aligned} S(\Phi_0) \equiv & [U(x) - 2\chi\rho_1] \Phi_0 - \\ & - i\hbar \left[f(k) \frac{\partial \Phi_0}{\partial k} + \varphi(k) \frac{\partial \Phi_0}{\partial \Theta} \right]. \end{aligned} \quad (4.10)$$

The condition that Φ_1 involves no secular terms is now reduced to the equality

$$\int S^*(\Phi_0) \frac{\partial \Phi_0}{\partial \xi} dx = 0,$$

from which taking into account (4.8) we obtain two equations

$$\varphi(k) = -\frac{1}{a\hbar} \int U(x) |\Phi_0|^2 dx, \quad (4.11)$$

$$f(k) = -\frac{2m}{(1+D)\hbar} \int |\Phi_0(x,t)|^2 U(x) \tanh[\eta(x-\xi)] dx, \quad (4.12)$$

where D is determined by the expression (3.16).

Substituting the value $f(k)$ obtained into Eq.(3.9b) and taking account the equality $m = m_{sol}(1+D)$ we find the equation

$$m_{sol} \frac{d^2 \xi}{dt^2} = F \quad (4.13)$$

determining the motion of the soliton centre.

The equation (4.13) resembles the Newton equation for a particle with a soliton mass moving under the influence of the force

$$F \equiv -2m\varepsilon \int U(x) |\Phi_0(x,t)|^2 \tanh[\mu(x-\xi)] dx. \quad (4.14)$$

Hence, the soliton motion is effected upon by the potential field $U(x)$ over the whole region where $|\Phi_0|^2 \neq 0$.

It is essential that the equation (4.13) involves an effective soliton mass rather than an exciton (electron) mass which enters directly in the nonlinear equation (3.2). This effect is due to the fact that in the molecular chains under consideration the nonlinearity is caused by the interaction of an exciton (electron) with the chain deformation (a phonon subsystem). The perturbation acts not only on an exciton (electrons) but also on a phonon subsystem which leads to the appearance of an additional term $\varepsilon \int_1 \rho(x,t)$ in the right-hand side of Eq. (4.6).

In a number of papers when studying the role of the perturbation $\varepsilon U(x)$ a nonlinear equation of the form

$$\left(i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + G |\Phi(x,t)|^2 \right) \Phi(x,t) = \varepsilon U(x) \Phi(x) \quad (4.15)$$

was considered without studying the nature of nonlinearity. In this case we obtain for the function $f(k)$ the value

$$f(k) = -\frac{2m}{\hbar} \int U(x) |\Phi_0(x,t)|^2 \tanh[\mu(x-\xi)] dx$$

which leads to the equation

$$m \frac{d^2 \xi}{dt^2} = F$$

for a quasi-particle with a mass m appearing in the equation (4.15).

Thus, in studying the dynamical behaviour of solitons described by a Schrodinger equation it is very important to know physical factors causing the equation nonlinearity.

5. The motion of a soliton in a one-dimensional molecular chain with allowance for thermal vibrations

In the above sections the assumption was made that the chain was

at a absolute zero temperature. It is interesting to investigate the influence of thermal vibrations of molecules on the soliton properties. This problem was considered in Ref. [9]. Below the basic results obtained in that paper are given.

To analyze the role of thermal motion it is necessary to express the displacements u_n in the peptide groups through the operators of creation b_q^+ and annihilation b_q of the phonons with wave numbers q

$$u_n = \sum_q \left(\frac{\hbar}{2MN\Omega_q} \right)^{1/2} (b_q + b_q^+) \exp(iqn a) \quad (5.1)$$

where $N (\gg 1)$ is the number of molecules in the chain

$$\Omega_q = |q| V_{aq}, \quad V_{aq} = a \sqrt{\alpha e/m}. \quad (5.2)$$

With temperature distinct from zero we should pass to the statistical description of quantum states of the system. The Hamiltonian function averaged by means of the density matrix of phonon states

$$\rho_{vv} = \frac{\langle v | \exp(-H_{ph}/\Theta) | v \rangle}{\sum_v \langle v | \exp(-H_{ph}/\Theta) | v \rangle}, \quad \Theta = kT, \quad (5.3)$$

takes the form

$$\begin{aligned} \mathcal{H} = \sum_n \left\{ \left(\frac{1}{2} - \frac{\hbar^2}{ma^2} \right) \varphi_n^* \varphi_n - \frac{\hbar^2 e^{-\bar{W}_n}}{2ma^2} \varphi_n^* (\varphi_{n-1} + \varphi_{n+1}) - \right. \\ \left. - \frac{1}{\sqrt{N}} \sum_q F(q) |\varphi_n|^2 (\beta_{qn} + \beta_{-q,n}) + \sum_q \hbar \Omega_q [\bar{\nu}_q + |\beta_{qn}|^2] \right\}, \quad (5.4) \end{aligned}$$

where

$$\begin{aligned} \bar{W}_n \approx \frac{a^2}{2} \sum_q (2\bar{\nu}_q + 1) q^2 |\beta_{qn}|^2, \\ F(q) \approx i2a\pi \frac{q}{|q|} \left(\frac{\hbar |q|}{2mV_{aq}} \right)^{1/2}, \quad (5.5) \end{aligned}$$

$$\bar{v}_q = \left[\exp \left(\frac{\hbar \Omega_q}{\Theta} \right) - 1 \right]^{-1}.$$

$\bar{v}_q$ is the average number of phonons in the state q ; $\beta_{qn}(t)$ determine the average values of displacements so that

$$\langle u_n(t) \rangle = - \sum_q \left(\frac{\hbar}{2MN\Omega_q} \right)^{1/2} \left[\beta_{qn}(t) + \beta_{-q,n}^*(t) \right].$$

The function (5.4) depends on φ_n , β_{qn} on canonically conjugate generalized momenta $i\hbar \varphi_n^*$ and $i\hbar \beta_{qn}^*$. Using this function we find the Hamiltonian equations which in the continuum approximation (for excitations moving with a velocity $V < V_{aq}$) are reduced to the following approximate equations

$$\left[i\hbar \frac{\partial}{\partial t} - \varepsilon_0 + \frac{\hbar^2}{2m(\Theta)} \frac{\partial^2}{\partial x^2} + G(\Theta) |\varphi(x,t)|^2 \right] \varphi(x,t) = 0, \quad (5.6)$$

$$\rho(x,t) = - \frac{2a\chi |\varphi(x,t)|^2}{M(1-s^2)V_{aq}^2}, \quad s = V/V_{aq}. \quad (5.7)$$

The Schrodinger equation (5.6) involves the parameters depending on temperature Θ .

The nonlinearity parameter is determined by the expression

$$G(\Theta) = \left[1 - \frac{1}{4} B F(\Theta) \right] G \quad (5.8)$$

where

$$G = 4a^2\chi^2 / M(1-s^2)V_{aq}^2, \quad (5.9)$$

$$B = \pi a^4\chi^2 / M(1-s^2)\hbar V_{aq}^3. \quad (5.10)$$

$$F(\theta) = \frac{a^2}{\pi^2} \int_0^{\pi/a} (1 + 2\sqrt{\frac{1}{2} - \frac{\theta^2}{\hbar^2 \Omega_0^2}}) q dq = \begin{cases} \frac{1}{2} + \frac{2\theta}{\hbar \Omega_0} & \text{if } \frac{\hbar \Omega_0}{\theta} \ll 1; \quad \Omega_0 = \pi V_{aq}/a; \\ \frac{1}{2} + \frac{\pi^2 \theta^2}{6 \hbar \Omega_0^2} & \text{if } \frac{\hbar \Omega_0}{\theta} \gg 1. \end{cases} \quad (5.11)$$

The effective exciton (electron) mass increases with increasing temperature by the exponential law

$$m(\theta) = m \exp \left(\frac{1}{4} \mu_0^2 a^2 B F(\theta) \right) \quad (5.12)$$

where

$$\mu_0 = 4 m a^3 \pi^2 / (2 M V_{aq}^2 \hbar^2 (1 - s^2)) \quad (5.13)$$

The increase in the effective exciton mass is due to the decrease in the resonance interaction between peptide groups caused by the phase coherence distortion of vibrational excitations in the neighbouring groups.

The nonlinearity parameter (5.8) decreases with increasing temperature. The nonlinearity parameter decrease is associated with the effective decrease in the connection of an exciton with the chain deformation. With sufficiently high temperature the parameter $G(\theta)$ may become negative. Then the equation (5.6) will have no localized (soliton) solutions. Unfortunately, this conclusion is only qualitatively meaningful since the theory is valid only when the inequality

$$B F(\theta) < 1 \quad (5.14)$$

is satisfied.

With the inequality $S < 1$ satisfied, the increase in the velocity of the excitation propagation according to (5.12), (5.10) and (5.13) leads to the increase in the values B and μ_0 causing the exciton mass increase since it acts as increasing temperature. The change of the nonlinearity parameter (5.8) is more complex. At $V \rightarrow V_{aq}$, according to (5.9), the value $G(\theta)$ increases while the multiplier $[1 - \frac{1}{4} B F(\theta)]$ decreases.

Solving the equation (5.6) we find that the probability of excita-

tion distribution along the chain is determined by the function

$$|\varphi(x,t)|^2 = \frac{a m(\theta)}{2 \cosh^2 \{m(\theta)[x-x_0-Vt]\}} \quad (5.15)$$

where

$$m(\theta) = a m(\theta) G(\theta) / 2\hbar^2. \quad (5.16)$$

In accordance with (5.7) the decrease in equilibrium distances between neighbouring molecules is determined by the function

$$\rho(x,t) = - \frac{2a^3 \chi m(\theta)}{M V_{aq}^2 \cosh^2 \{m(\theta)[x-x_0-Vt]\}} \quad (5.17)$$

The energy of a soliton traveling with velocity $V \ll V_{aq}$ is proportional to the square of velocity

$$E_{sol}(V) = E_{sol}(0) + \frac{1}{2} m_{sol} V^2, \quad (5.18)$$

with the soliton mass

$$m_{sol} = m(\theta) \left[1 + \frac{a^2 G^2(\theta)}{6 \hbar^2 V_{aq}^2} \right] \quad (5.19)$$

and the energy at rest

$$E_{sol}(0) = \mathcal{E}_0 - m(\theta) a^2 G^2(\theta) / 12 \hbar^2. \quad (5.20)$$

The soliton size is determined by the value

$$l \cong \frac{\pi}{m(\theta)} = 2\pi \hbar^2 / a m(\theta) G(\theta). \quad (5.21)$$

We substitute the values (5.8) and (5.12) into this expression.

Then when the continuum approximation $a m_0 \ll 1$ and inequality (5.14) are satisfied

$$l \approx 2\pi \hbar^2 / (m a [1 - \frac{1}{4} B F(\theta)] G). \quad (5.22)$$

Thus, with increasing temperature the soliton size increases and its properties more and more approximate the exciton properties. The fact whether a soliton transforms into an exciton continuously or discontinuously can be clarified only on the basis of a new theory valid in the region $V \approx V_{aq}$.

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